

Formulas

Description_length

$$\text{If} \left(\begin{array}{l} \text{Length} \left(\text{Trim} \left(\text{description}^{\wedge} \right) \right) == 0 \Rightarrow 0 \\ \text{else} \Rightarrow \left(1 + \text{Length} \left(\text{description} \right) \right) - \text{Length} \left(\text{Substitute} \left(\text{description}, "", ""^{\wedge} \right) \right) \end{array} \right)$$

Listing_price_imp

$$\text{If} \left(\begin{array}{l} \text{Is Missing} \left(\text{listing_price} \right) \Rightarrow \frac{\text{sale_price}}{\left(1 - \text{discount} \right)} \\ \text{else} \Rightarrow \text{listing_price} \end{array} \right)$$

Units_sold

$$\frac{\text{revenue}}{\text{sale_price}}$$

Listing_price_quartile

$$\text{If} \left(\begin{array}{l} \text{listing_price_imp} \leq 49.99 \Rightarrow 1 \\ \text{else} \Rightarrow \text{If} \left(\begin{array}{l} \text{listing_price_imp} \leq 75.99 \Rightarrow 2 \\ \text{else} \Rightarrow \text{If} \left(\begin{array}{l} \text{listing_price_imp} \leq 109.95 \Rightarrow 3 \\ \text{else} \Rightarrow 4 \end{array} \right) \end{array} \right) \end{array} \right)$$

Description_length_quartile

$$\text{If} \left(\begin{array}{l} \text{description_length} \leq 34 \Rightarrow 1 \\ \text{else} \Rightarrow \text{If} \left(\begin{array}{l} \text{description_length} \leq 41 \Rightarrow 2 \\ \text{else} \Rightarrow \text{If} \left(\begin{array}{l} \text{description_length} \leq 48 \Rightarrow 3 \\ \text{else} \Rightarrow 4 \end{array} \right) \end{array} \right) \end{array} \right)$$

Low_revenue_flag

$$\text{If} \left(\begin{array}{l} \text{revenue} \leq 885.28 \Rightarrow 1 \\ \text{else} \Rightarrow 0 \end{array} \right)$$

Analysis

Figure 1. Relationship Between Reviews and Estimated Units Sold

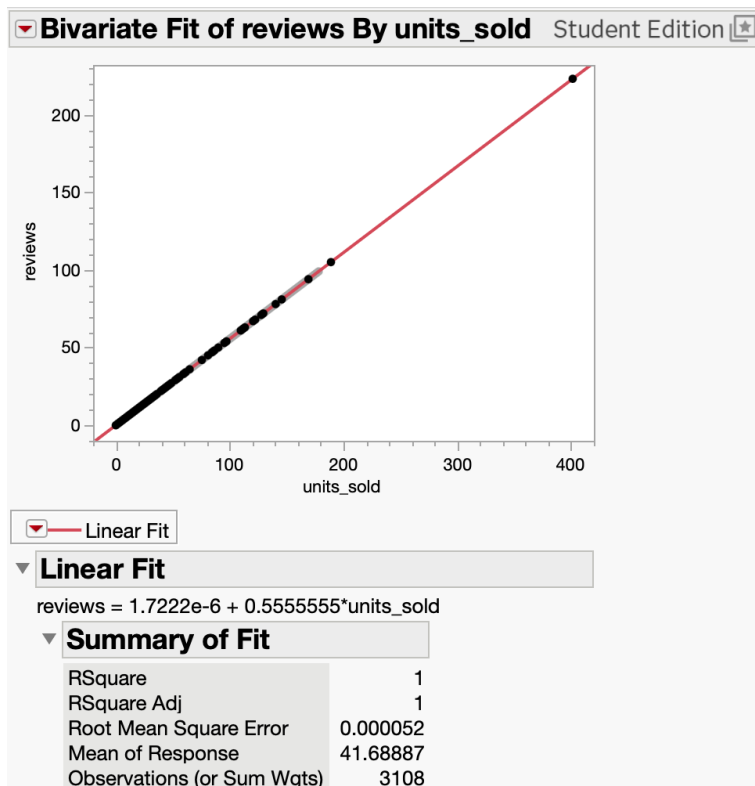


Figure 2. Scatterplot Matrix and Correlations Among Continuous Predictor Variables

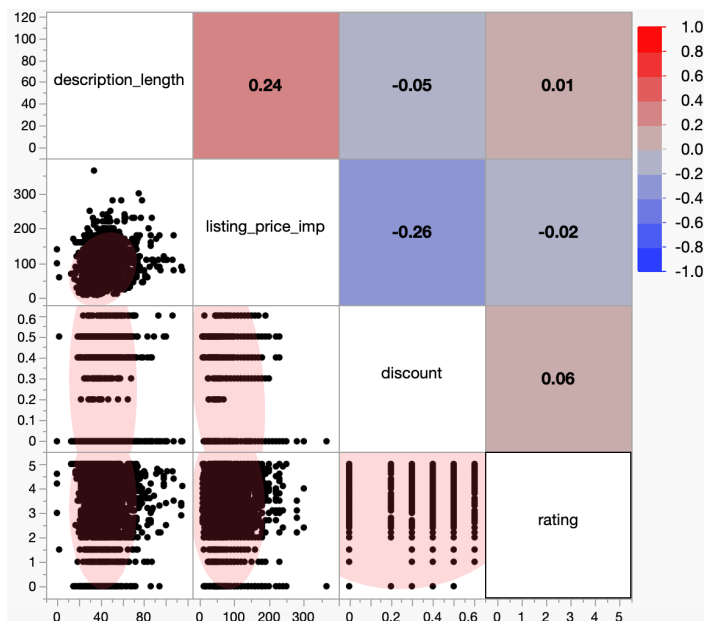
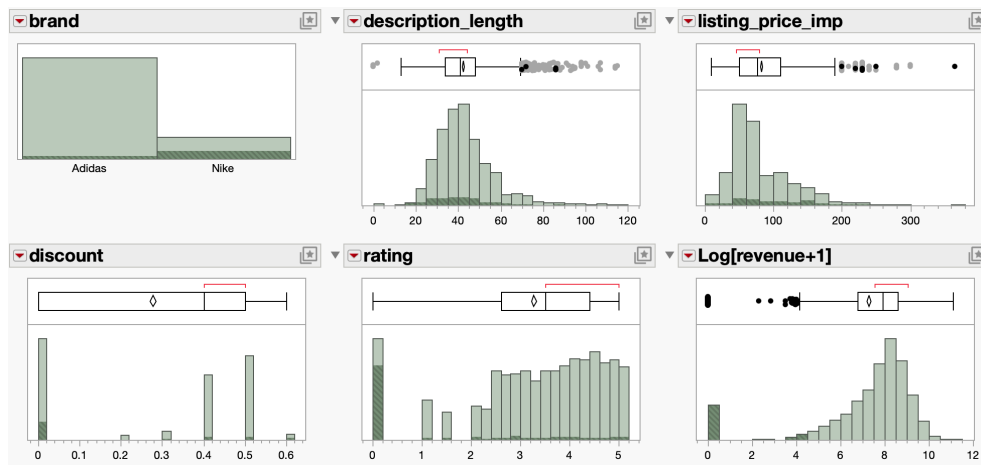


Figure 3

(A) Low-Revenue Observations Across Predictor Distributions



(B) Lower Listing Price Segment



(C) Zero-Discount Observations



Figure 4. Average Revenue Across Listing Price Quartiles for Adidas and Nike Products

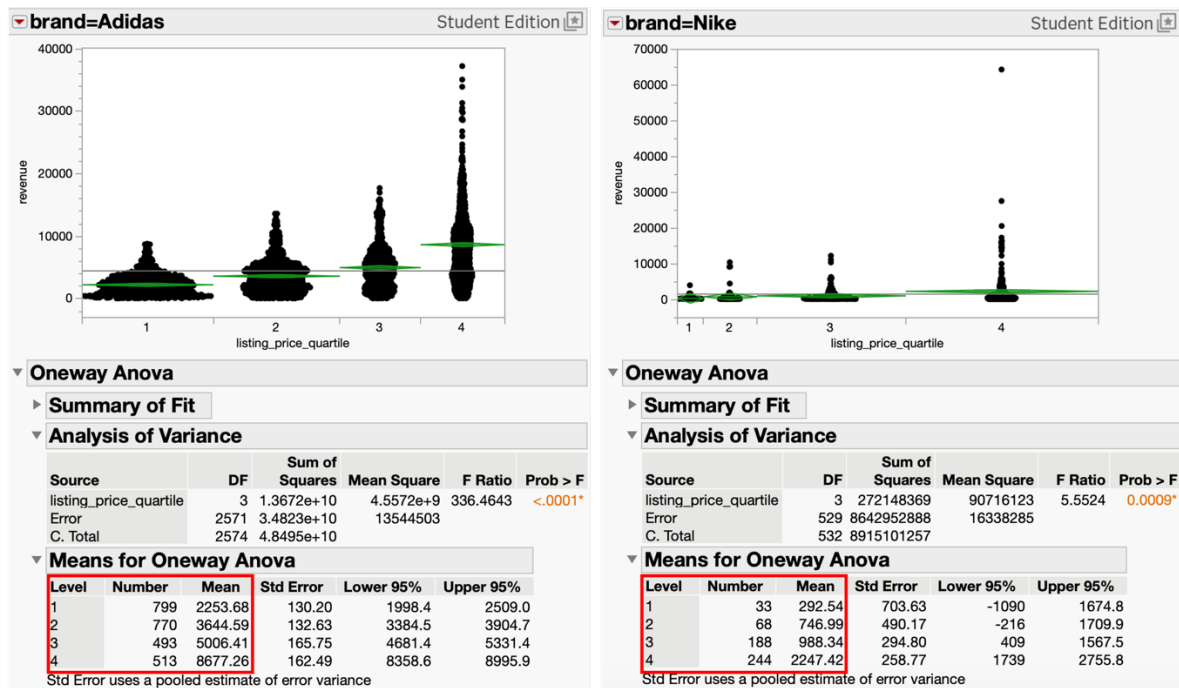


Table 1

Average Revenue and Units Sold by Brand and Listing Price Quartile

Brand	Quartile	Mean Revenue	Units Sold
Adidas	Q1	2253	70072
	Q2	3645	67255
	Q3	5006	43929
	Q4	8677	44741
Nike	Q1	293	308
	Q2	747	765
	Q3	988	2182
	Q4	2247	3973

Figure 5. Product Rating by Description Length Quartile for Adidas and Nike Products

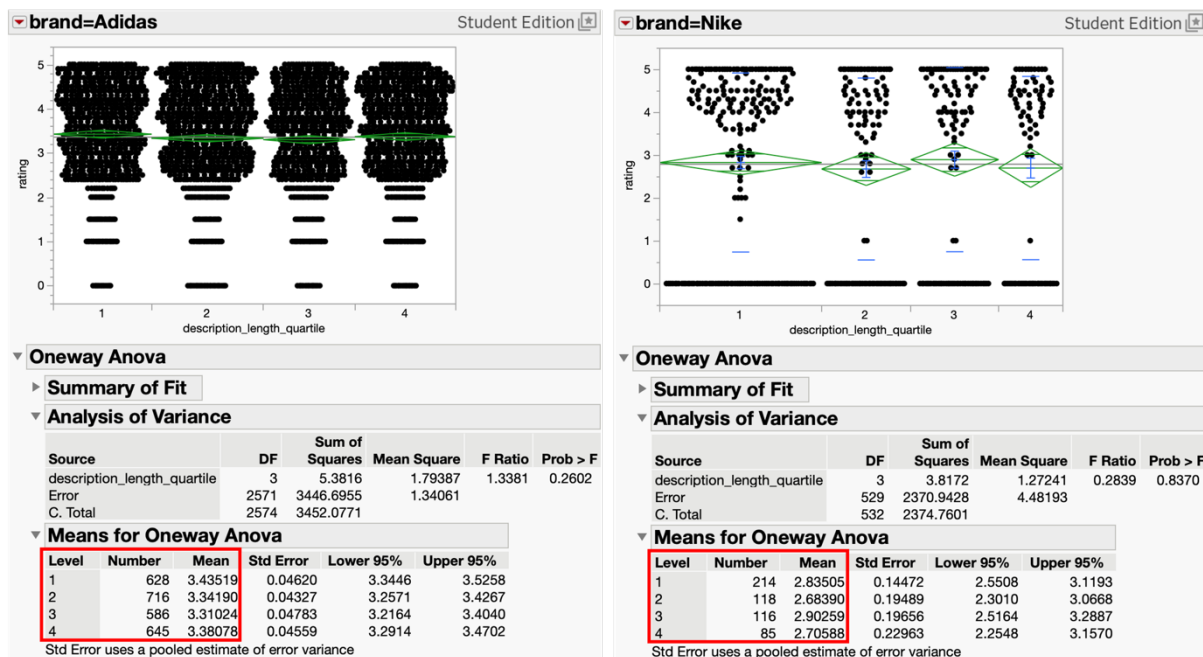


Figure 6. Data Partition for Training, Validation, and Test Sets

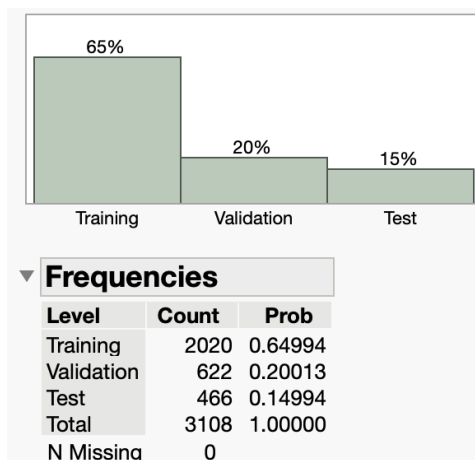


Figure 7. Comparison of Predictive Model Performance on the Test Set

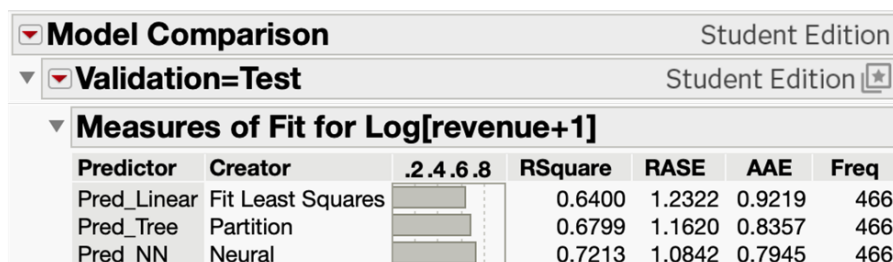


Figure 8. Linear Regression Model Performance and Predictor Importance.

(A) Actual vs. predicted values for the validation set.

(B) Standardized regression coefficients showing predictor importance.

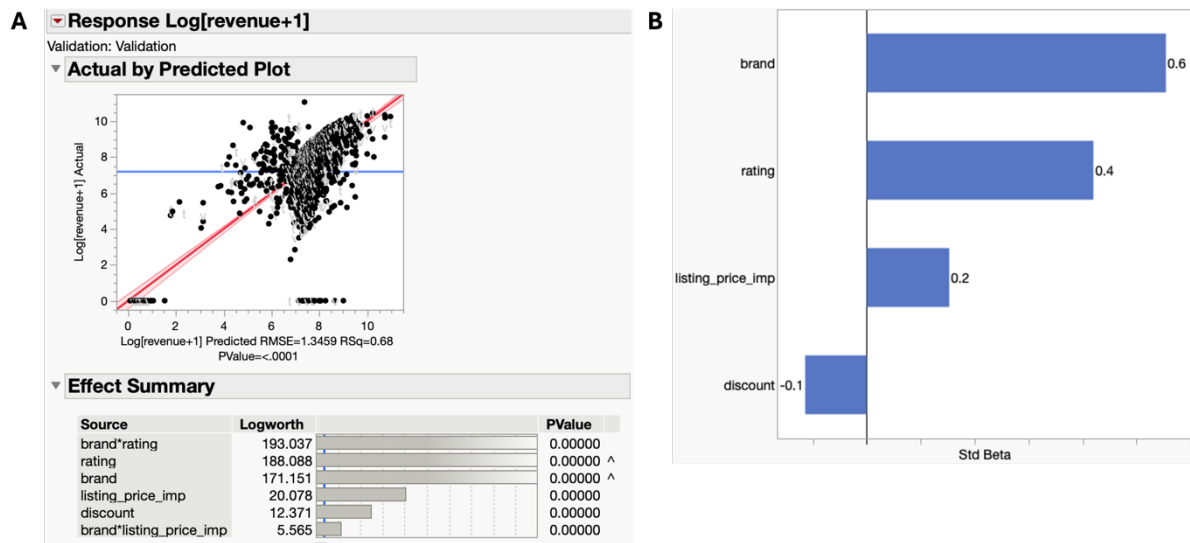


Figure 9. Regression Tree Model for Log(Revenue + 1).

(A) Regression Tree Structure.

(B) Variable Importance Based on Independent Resampled Inputs.

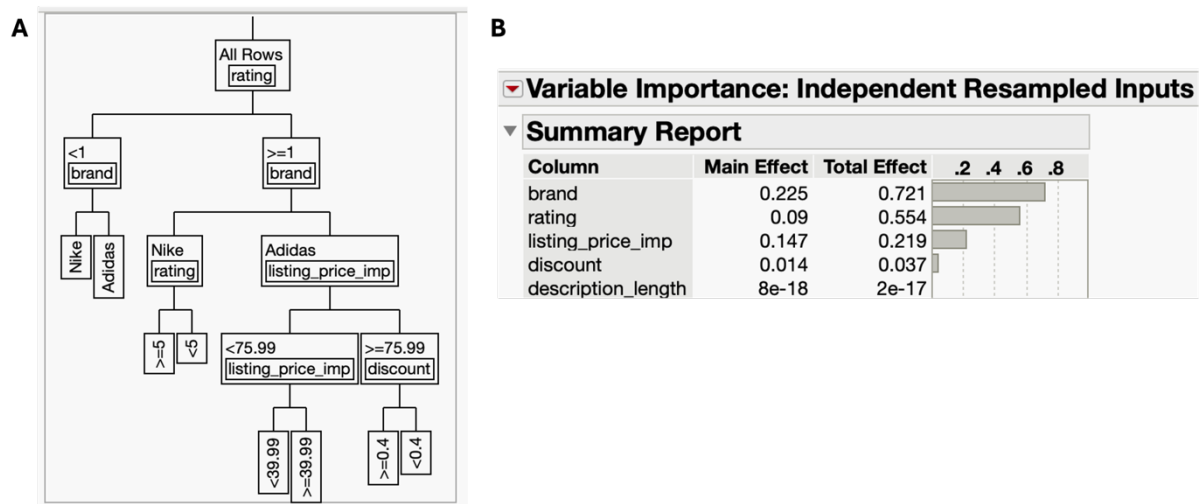


Figure 10. Neural Network Model Results for Log(Revenue + 1).
(A) Network Architecture with One Hidden Layer and Five Neurons.
(B) SHAP-Based Predictor Importance.

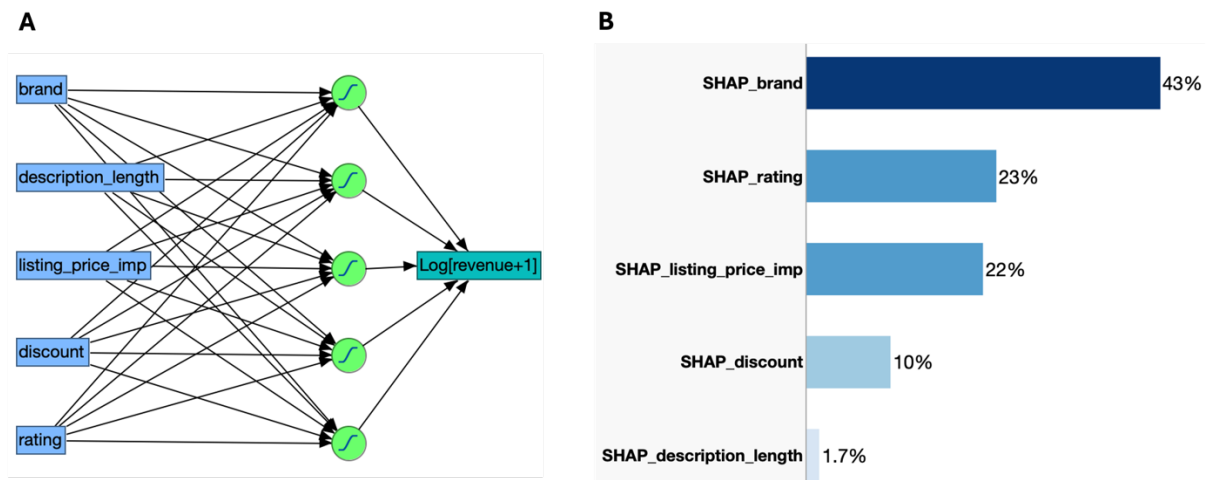


Figure 11. Brand–Rating Interaction Across Different Modeling Approaches.
(A) Linear Regression Model.
(B) Regression Tree Model.
(C) Neural Network Model.

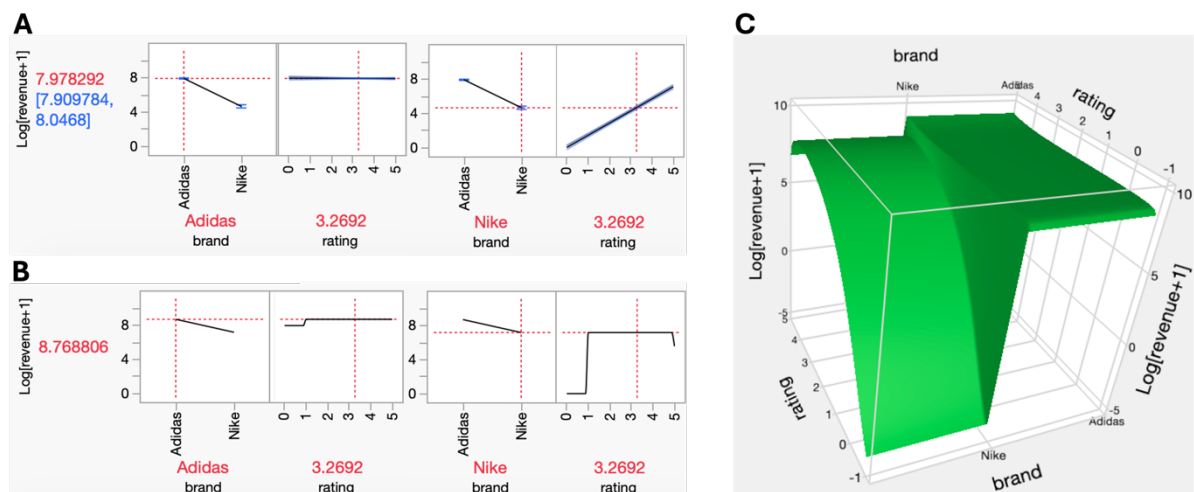


Figure 12. Distribution of the Low-Revenue Flag Across Training, Validation, and Test Sets

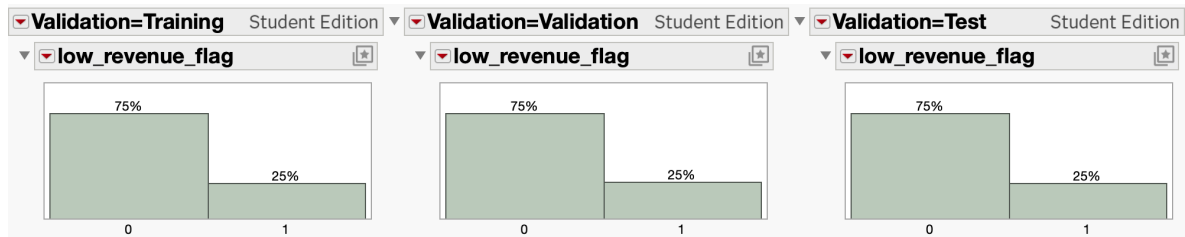


Figure 13. Comparison of Classification Models for Low-Revenue Prediction (ROC Curves and Performance Metrics, Test Set)

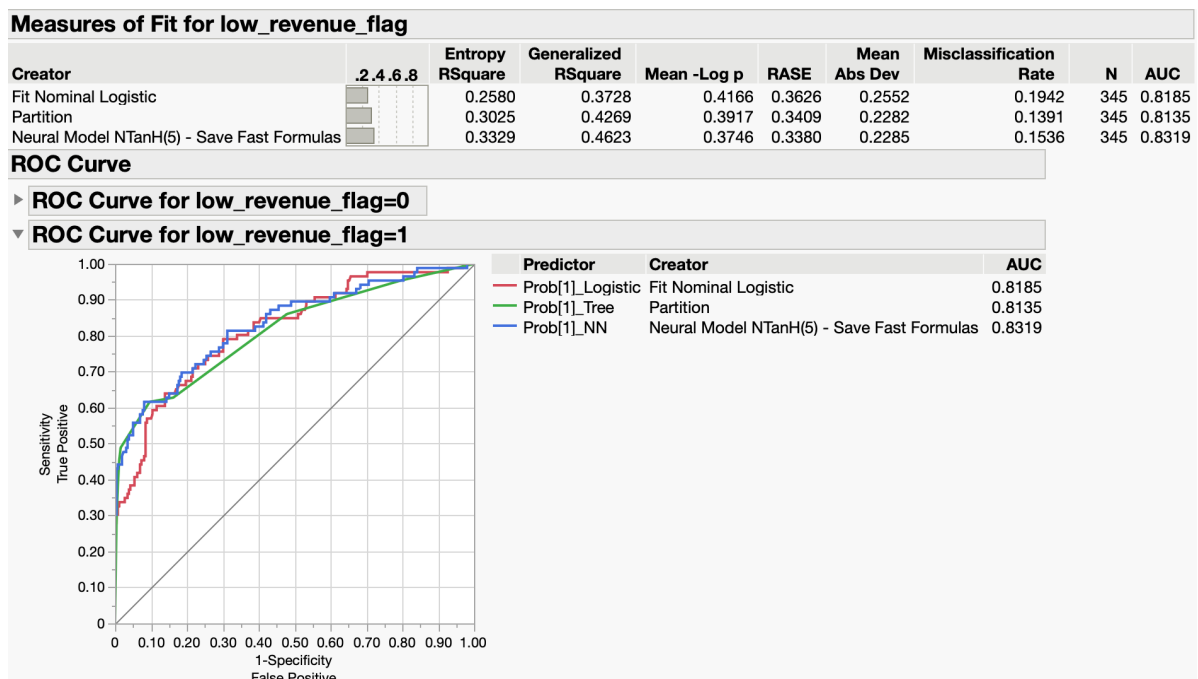


Table 2

Model Performance Across Classification Thresholds

Threshold	Best Sensitivity	Best Precision	Best F1
0.50	0.4884	0.913	0.6364
0.45	0.5233	0.913	0.6364
0.40	0.6163	0.7273	0.6463
0.35	0.6163	0.7067	0.6584
0.30	0.6163	0.6795	0.6463
0.25	0.6744	0.5625	0.5949

Logistic Regression

Decision Tree

Neural Networks

Figure 14. Classification Outcomes at Probability Threshold 0.35

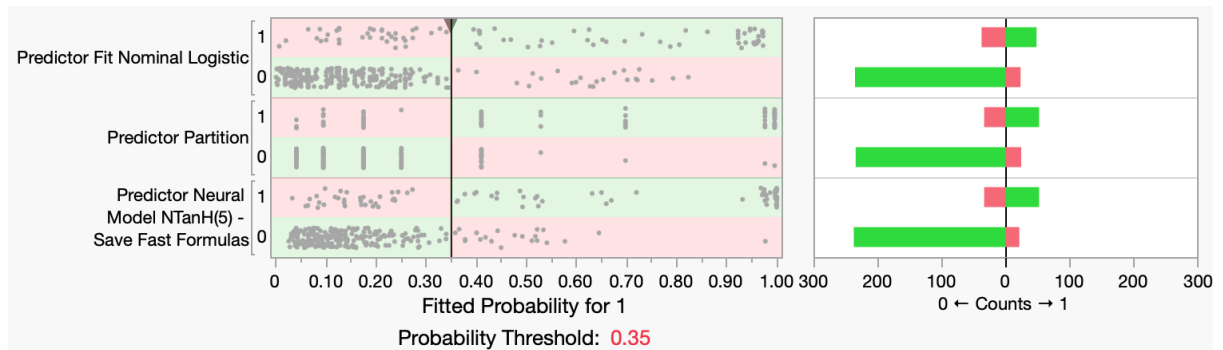


Figure 15. Decision Tree for Predicting Low-Revenue Products

